1 Solution — GIAM 3.1 Problem 1

1.1 Problem

Every prime number greater than 3 is of one of the two forms $6k+1$ or $6k+5$. **What statement(s) could be used as hypotheses in proving this theorem?**

1.2 Answer

$$\boxed{\begin{array}{ll} \textbf{1.} & p \text{ is prime} \\ \textbf{2.} & p > 3 \end{array}}$$

These are the two hypotheses. To find them, rewrite the theorem as an explicit universally quantified conditional:

$$\forall p \in \mathbb{Z}, \quad \underbrace{\big(p \text{ is prime}\big) \;\wedge\; \big(p > 3\big)}_{\text{hypotheses}} \implies \underbrace{\exists k \in \mathbb{Z},\; \big(p = 6k+1\big) \vee \big(p = 6k+5\big)}_{\text{conclusion}}.$$

Everything to the left of the arrow is what a proof may **assume**; everything to the right is what it must **establish**.

1.3 Finding the Hypotheses

Rewrite the theorem as a conditional. Whatever sits to the left of the arrow is what you get to assume.

as stated:	"Every prime number greater than 3 is of the form 6k+1 or 6k+5."
rewritten:	For all integers p, if p is prime **and** p > 3, then p = 6k+1 or p = 6k+5 for some integer k. *the "every ... is" phrasing hides both a ∀ and an ⟹*

HYPOTHESES — what you assume

1. p is prime
2. p > 3

both are needed — see below

CONCLUSION — what you must reach

$\exists k \in \mathbb{Z}$ such that

p = 6k+1 or p = 6k+5

One more ingredient, not a hypothesis but a known theorem you may cite:

the **quotient–remainder theorem** — every integer p is p = 6k + r with r ∈ {0, 1, 2, 3, 4, 5}.

How the hypotheses get used: six cases, four eliminated

r	p = 6k + r	what rules it out	uses
0	6k	even, and p > 2, so composite	both
1	**6k+1**	**nothing — survives**	
2	6k+2	even, and p > 2, so composite	both
3	6k+3 = 3(2k+1)	a multiple of 3, and p > 3, so composite	both
4	6k+4	even, and p > 2, so composite	both
5	**6k+5**	**nothing — survives**	

Every elimination needs **both** hypotheses: "prime" to say a factorisation is fatal, "p > 3" to rule out p = 2 and p = 3.

Checked: all 301 primes between 3 and 2000 are 1 or 5 mod 6, and both forms occur.

Figure 1.1: Top: the theorem as stated, 'Every prime number greater than 3 is of the form 6k+1 or 6k+5', rewritten as: for all integers p, if p is prime and p is greater than 3, then p equals 6k+1 or p equals 6k+5 for some integer k — noting that the 'every ... is' phrasing hides both a for-all and an implication. Two boxes follow. HYPOTHESES, what you assume: p is prime, and p greater than 3 — both are needed. CONCLUSION, what you must reach: there exists an integer k such that p equals 6k+1 or p equals 6k+5. A further panel notes one more ingredient that is not a hypothesis but a citable theorem: the quotient-remainder theorem, that every integer p is 6k+r with r in zero through five. Finally a six-row table of cases. Residue 0, p equals 6k: even and p exceeds 2, so composite. Residue 1, highlighted: nothing rules it out, survives. Residue 2: even, composite. Residue 3, p equals 3 times (2k+1): a multiple of 3 with p exceeding 3, so composite. Residue 4: even, composite. Residue 5, highlighted: survives. Every elimination uses both hypotheses. Checked: all 301 primes between 3 and 2000 are 1 or 5 mod 6, and both forms occur.

The English phrasing "**every** prime number **greater than 3** *is* of the form..." conceals both a quantifier and an implication. Unpacking it is the whole exercise:

- "every" $\longrightarrow$ $\forall p$, and the proof will open "let p be an arbitrary..."
- "prime number greater than **3**" $\longrightarrow$ *two* conditions, joined by **and**

- "is of the form..." $\longrightarrow$ the conclusion, itself an existential ($\exists k$) inside a disjunction

1.4 How the Hypotheses Are Used

A proof would also cite the **quotient–remainder theorem** — every integer can be written $p = 6k + r$ with $r \in \{0, 1, 2, 3, 4, 5\}$ — and then eliminate four of the six cases:

r	$p = 6k + r$	**why it is impossible**	**uses**
0	$6k$	even, and $p > 2$, so composite	both
1	$6k + 1$	**survives**	
2	$6k + 2$	even, and $p > 2$, so composite	both
3	$6k + 3 = 3(2k + 1)$	a multiple of 3, and $p > 3$, so composite	both
4	$6k + 4$	even, and $p > 2$, so composite	both
5	$6k + 5$	**survives**	

Table 1.1

Only $r = 1$ and $r = 5$ remain, which is the conclusion. Verified computationally: all 301 primes strictly between 3 and 2000 are 1 or 5 mod 6, and both forms genuinely occur.

1.5 Remark — Both Hypotheses Are Genuinely Needed

It is worth checking that neither hypothesis is decorative, since a well-posed theorem should not carry idle assumptions.

Drop "prime." Then the claim fails immediately: $4 = 6 \cdot 0 + 4$ and $9 = 6 \cdot 1 + 3$ are greater than 3 but of neither form. Without primality there is nothing to rule out the even residues or $r = 3$.

Drop "$p > 3$." Then the two small primes break it: $2 = 6 \cdot 0 + 2$ and $3 = 6 \cdot 0 + 3$. These are exactly the primes that *divide* 6, and they are the reason the bound is stated at all — the same phenomenon noted in Problem 2.5.4, where 2 and 3 were likewise the only exceptions to a mod-6 claim.

So the theorem is stated with the minimum needed, and each elimination in the table above draws on both assumptions at once: primality makes a factorisation fatal, and $p > 3$ guarantees the factorisation is proper.

1.6 Remark — Hypotheses vs. Facts You May Cite

A subtlety worth separating. The quotient–remainder theorem is *used* in the proof but is **not** a hypothesis of this theorem. The distinction:

- A **hypothesis** is an assumption about the object under discussion — here, about p. It is discharged at the end, and it is what makes the result conditional.
- A **previously proved theorem** is available unconditionally. Citing it costs nothing and adds nothing to the statement being proved.

Confusing the two leads to theorems cluttered with assumptions that were never needed. The test is whether removing the statement would change *what is being claimed*: dropping "p is prime" changes the theorem; dropping the quotient–remainder theorem merely leaves you unable to prove it.

1.7 Remark — The Conclusion Is a Disjunction, Which Shapes the Proof

Notice the *form* of what must be established: an existential wrapped around a disjunction, $\exists k,\ (p = 6k + 1) \vee (p = 6k + 5)$. That shape suggests the strategy.

Because the conclusion is a disjunction, one does not have to decide *which* form holds — it suffices to rule out every alternative, which is why the proof proceeds by elimination rather than by construction. This is the **extended elimination** rule of Problem 2.7.2 in action: from $r \in \{0, 1, 2, 3, 4, 5\}$ (a six-fold disjunction) together with four negations, conclude $r = 1$ or $r = 5$. Reading the conclusion's logical shape before starting is often the fastest way to pick a proof technique.

1.8 Remark — Why 6 and Not Some Other Modulus

The modulus $6 = 2 \cdot 3$ is not arbitrary. Working mod 6 lets a single case analysis dispose of divisibility by *both* small primes at once: residues $0, 2, 4$ die by 2, and residue 3 dies by 3. Two of the six residues survive, so roughly a third of the residue classes can contain large primes.

Pushing further, one could work mod $30 = 2 \cdot 3 \cdot 5$, where eight residues survive ($1, 7, 11, 13, 17, 19, 23, 29$) — the basis of "wheel factorisation," a standard speed-up for prime sieves. The general statement is that a prime p exceeding all prime factors of n must be coprime to n, so it lands in one of the $\phi(n)$ invertible residue classes. Here $\phi(6) = 2$, which is exactly the two forms in the theorem.